

SELENA DUNCAN

ABOUT ME

14 years in Data/AI/ML, most of it spent on systems other teams depend on. Comfortable leading technical decisions, pushing back on scope when it matters, and carrying a pager for what I build. Primary tools: Hugging Face and Docker.

EXPERIENCE

AI Researcher, Shopify

2015–2026

- Reduced label noise by redesigning the annotation guidelines, improving validation accuracy by 40%.
- Ran A/B tests across 3 model variants; the winning variant added an estimated \$30k in annual revenue.
- Productionised demand forecasting models that reduced stock-outs by 60% across 3 regions.
- Trained and deployed Scikit-learn models serving 5M predictions per day at under 180ms p95.
- Built the feature store that 12 downstream models now read from, ending duplicated feature logic.

Lumen Works — Applied Scientist

2013–2015

- Cut inference cost by 18% through quantisation, batching and autoscaling on AWS.
- Translated ambiguous business questions into a modelling roadmap with product and finance stakeholders.
- Lifted recommendation click-through by 20% through feature engineering and retraining cadence in LangChain.
- Mentored 10 analysts moving into modelling work, with weekly paper reading and code review.

Junior MLOps Engineer, BrightPath Solutions

2012–2013

- Developed an NLP classification system in Hugging Face reaching 92% F1, replacing a manual review queue.
- Built an end-to-end training pipeline on Pandas, cutting retraining from days to 10 hours.
- Migrated ad-hoc notebooks into versioned Hugging Face pipelines with reproducible runs and tracked experiments.

Machine Learning Engineer, Adobe

2011–2012

- Established offline/online evaluation parity, catching 5 silent regressions before release.
- Wrote the team's model documentation and review standard, now required before any production launch.
- Represented the team in cross-functional planning with product, design and support.

ACADEMIC BACKGROUND

Bachelor, Physics

Universidad Politecnica de Madrid | 2008–2011

TECH STACK

Languages: Python, Ruby

Frameworks: Hugging Face, Pandas, LangChain, Scikit-learn

Cloud & DevOps: Docker, AWS

Data: Airflow, BigQuery, DynamoDB

Testing & Tooling: Postman, Playwright

SELECTED PROJECTS

Technical blog

Writing up Scikit-learn experiments and failures, roughly 8k readers a month.

Home energy dashboard

Collects meter readings and renders them with DynamoDB. Mostly an excuse to learn time-series storage properly.

TRAINING & CERTIFICATIONS

Certified Kubernetes Administrator (CKA) — Cloud Native Computing Foundation, 2017

VOLUNTEERING

STEM ambassador running school workshops twice a term.

INTERESTS

Jazz guitar, Home automation